Mp10 AutoTasks — Administrator's Guide

Backups, electronic remittances, modality worklist, and automated reports

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1 About this guide

Mp10 includes a background service called **AutoTasks**. It runs unattended on your server and takes care of work that would otherwise have to be done by hand every day:

- taking a **backup** of your data on a schedule you choose, archiving it and checking the archive is sound,
- **collecting files from your clearinghouse** — remittances, claim status responses and acknowledgements — and posting them into Mp10,
- reading **remittance PDFs** downloaded from insurer portals,
- asking insurers whether a patient's **coverage is active**,
- texting patients an **appointment reminder** they can reply to,
- keeping a **modality worklist** in step with the schedule, so imaging equipment can be told who its next patient is,
- keeping the **CMS reference data** behind the claim checks current — the Medicare fee schedule and NCCI tables each quarter, the ICD-10-CM code set each year — and e-mailing what it did, and
- running **reports** on a schedule and e-mailing them to whoever needs them.

Each of those is turned on separately. Nothing in this list happens until you set the corresponding entry in the System Registry, with two exceptions noted in their own chapters: eligibility requests and the CMS reference updates default to on.

This guide is for the person who administers Mp10 at the practice. It does not assume you are a programmer. It does assume you can open the **Admin10** module and that you know where your practice keeps its shared folders.

Anything in this guide that must be done by whoever installed Mp10 (your IT person or your reseller) is marked **IT task**.

> **A note on wording.** Throughout this guide, *setting* means an entry in the Mp10 System Registry — not the Windows registry. They are unrelated. Nothing in this guide requires you to open Windows' own registry editor.

2 How AutoTasks works

AutoTasks runs as a Windows service on your Mp10 server. Once started it wakes up at a fixed interval, checks whether there is anything to do, does it, and goes back to sleep.

Setting	Section	Default	What it does
`RUN_EVERY_N_SECONDS`	`AUTOTASKS`	`120`	How often the service wakes up, in seconds

Two consequences worth understanding, because they explain most "why hasn't it happened yet?" questions:

- **Nothing happens instantly.** If you drop a file into a folder, AutoTasks picks it up on its next wake-up — up to two minutes later by default.
- **Scheduled work fires on the first wake-up at or after the time you set**, not exactly on the second.

2.1 The log file

AutoTasks writes everything it does to a log file sitting next to the service program, named after it with a `.out` extension. On a typical install:

`C:\Mp10\AutoTasks.out`

This file is the single most useful thing to look at when something has not happened. It is plain text — open it with Notepad. Every line is stamped with the date and time.

The log is capped so it cannot fill the disk:

Setting	Section	Default	What it does
`MAXIMUM_LOG_FILE_SIZE`	`GENERAL`	`8192`	Largest log size in KB before it is trimmed

3 Where settings live: the System Registry

All three capabilities in this guide are switched on and configured in one place: **Admin10 → Global System Registry**.

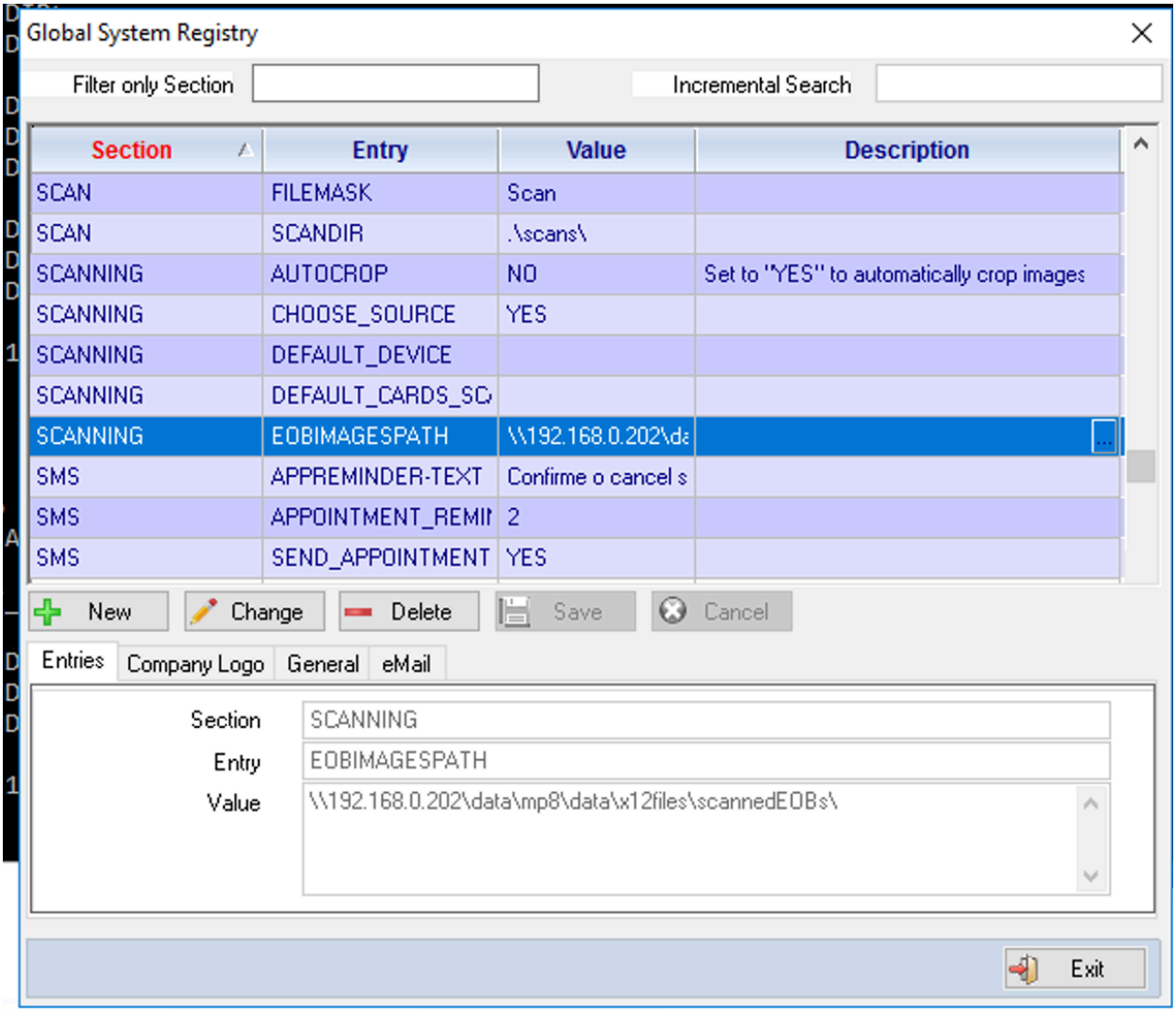


Figure 1: The Global System Registry in Admin10

Every setting has four parts:

Column	Meaning
Section	The group a setting belongs to, e.g. `BACKUPS`, `SCANNING`
Entry	The name of the setting, e.g. `BACKUP_PATH`
Value	What you are setting it to
Description	A free-text note for whoever reads it next

3.1 Adding or changing a setting

1. Open **Admin10** and go to the **Global System Registry**.

2. To find an existing setting, type into **Filter only Section** (for example `BACKUPS`) or use **Incremental Search**.
3. Press **New** to add a setting, or select a row and press **Change** to edit one.
4. Fill in **Section**, **Entry** and **Value** in the *Entries* tab at the bottom.
5. Press **Save**.

   Type the Section and Entry names exactly as shown in this guide. They are matched by name. A misspelled entry is not an error — it is simply a setting nobody reads, so the feature stays switched off with no warning.

AutoTasks re-reads its settings at the start of every cycle, so a change you make in the System Registry takes effect within a couple of minutes. There is nothing to restart.

The exception is anything the service holds in memory rather than reads: the daily jobs remember that they have already run today, so a backup that has already failed today will not try again until tomorrow unless the service is restarted (see *Backups*).

4 Backups

AutoTasks can take an **Advantage Database Server hot backup** — a consistent copy of your live data taken while people are still working in Mp10 — and optionally compress it into a dated zip file, keeping only the most recent few.

4.1 Turning backups on

Backups run only when **all three** of these are set. Leave any one of them blank and no backup will ever run.

Entry	Example	What it means
`BACKUP_PATH`	`\\adsserver\backup\mp10\`	Where the backup image is written
`BACKUP_TIMES`	`02:00:00`	Earliest time of day the backup may start
`BACKUP_DAYS`	`2345`	Which days of the week it may run

All three go in section `BACKUPS`.

One optional setting lives with them:

Entry	Example	Default	What it means
`BACKUP_MAX_HOURS`	`8`	`8`	How long the database backup may run before AutoTasks is treated as hung and restarted

AutoTasks watches itself for hangs: if it stops responding, it restarts rather than sitting there doing nothing all night. The database backup is the one step that legitimately runs for hours, so it gets its own allowance, and `BACKUP_MAX_HOURS` is it. Set it comfortably above the longest backup your site has ever taken — the completion e-mail tells you what that is.

> **If a backup is cut short you are told about it.** AutoTasks notices at its next start that a backup was running when it stopped, and e-mails `Mp10 backup INTERRUPTED`. See *Being told what happened* below.

4.1.1 BACKUP_PATH — read this carefully

This is the one setting people most often get wrong.

The backup is written by the **Advantage Database Server**, not by Mp10 and not by your workstation. The path must therefore make sense *to the database server process*:

- **UNC paths work** — `\\server\share\folder\`
- **Local drive letters on the database server work** — `D:\Backups\`
- **Mapped network drives do NOT work** — `Z:\Backups\` will fail, even if `Z:` is perfectly visible from your own PC

The account the Advantage service runs under must have permission to write there. *(IT task.)*

4.1.2 BACKUP_DAYS — the day codes

Days are given as a string of digits run together, one digit per allowed day:

Digit	Day
1	Sunday
2	Monday
3	Tuesday
4	Wednesday
5	Thursday
6	Friday
7	Saturday

So:

Value	Meaning
`2345`	Monday through Thursday
`234567`	Monday through Saturday
`1234567`	Every day
`7`	Saturdays only

4.1.3 BACKUP_TIMES

A 24-hour time, `HH:MM:SS`. `02:00:00` means "any time from 2 a.m. onward".

Pick a time when the practice is quiet. A hot backup does not lock anybody out, but it does work the server hard.

A backup is attempted at most once per calendar day. Once AutoTasks has started one, it will not start another today, however many times it wakes up.

> **A failed attempt still counts as the day's attempt.** If the backup fails, AutoTasks does not keep retrying — it waits until tomorrow. This is deliberate: a hot backup runs for hours and hammering the database server all day cannot fix a wrong path or a full disk. It does mean **a failure needs somebody to act on it**, which is what the failure e-mail is for. If you fix the cause and want a backup the same day, stop the AutoTasks service, delete that dictionary's `backup_lastattempt_….txt` next to the AutoTasks program (see below), and start the service again. A restart on its own is not enough: the day's attempt is remembered on disk so that a restart cannot set off a second hours-long backup.

One AutoTasks serving several dictionaries. When `Mp10.ini` lists more than one (`PATH01`, `PATH02`, … under `[RDD]`), each dictionary is backed up on its own schedule, from its own `BACKUPS` entries, and reported from its own `eMail` settings. The once-a-day rule applies to each dictionary separately: one dictionary's backup never uses up another's day. The day's attempt is remembered in a file per dictionary next to the AutoTasks program, named after the dictionary's path — for `\\server\e$\AdsData\Ver10\mp.add` it is `backup_lastattempt_SERVER_E_ADSDATA_VER10_MP_ADD.txt`.

> **A dictionary with no outgoing mail server set up is backed up silently.** The notices go out through the `eMail` section of the dictionary being backed up, not through another dictionary's. If one dictionary's notices never arrive, check that its `eMail` section has `SMTP-ServerName`, `SMTP-User`, `SMTP-Pass` and `Address` filled in — the AutoTasks log says "No outgoing mail server is configured" when they are not.

4.2 Optional: zipping and retention

If you also set a zip path, AutoTasks compresses each backup into a dated archive, **reads that archive back to check it**, and only then prunes old ones.

Entry	Example	Default	What it means
`BACKUP_ZIP_PATH`	`D:\Backups\Zips\`	*(blank)*	Where dated zips are written. Blank = do not zip
`MAX_BACKUPS_TO_KEEP`	`20`	`20`	How many zips to keep. Oldest beyond this are deleted
`ARCHIVE_MAX_HOURS`	`12`	`12`	How long the zip step may run before it is stopped and reported as failed
`FILES_TO_SKIP`	`oldtable.adt;temp.adt`	*(blank)*	Data files to leave out of the backup, separated by semicolons

Zip files are named by date, so they sort chronologically:

`Mp10Data-2026-07-31.zip`

Unlike `BACKUP_PATH`, the zip path is used by AutoTasks itself, so an ordinary path reachable from the Mp10 server is fine.

> **`MAX_BACKUPS_TO_KEEP` deletes files.** Once the number of zips in `BACKUP_ZIP_PATH` exceeds this number, the oldest are erased automatically. Keep that folder for backups only, and set the number to match the disk space you have.

4.2.1 Files that cannot be compressed are copied instead

Not all data compresses. A memo file holding scanned documents is already compressed — the scans are JPEGs or PDFs — and running it through compression again reads every byte for nothing, making the result very slightly *larger*.

AutoTasks samples each file before adding it, and simply copies the ones compression cannot help. The archive comes out the same size either way; the difference is that no time is wasted on files that were never going to shrink.

> **A practice storing images in Mp10 should not expect a small archive.** If most of your data is scanned documents, the zip will be close to the size of the backup itself. That is the data, not a fault — and it means the archive folder needs roughly as much free space as the backup folder, for **each** archive you keep. Check `MAX_BACKUPS_TO_KEEP` against the free space you actually have.

4.2.2 The archive is checked before it counts as a backup

An archive is not treated as a backup until AutoTasks has opened it again and confirmed every file inside it is intact and the right size. Only then does it delete anything older.

This matters more than it sounds. If the archive cannot be written or does not check out, **nothing is deleted** — the previous archives are left exactly as they were, and you get a failure e-mail. Older backups are never traded away for a new one that might not work.

What happens to the failed archive depends on how it failed, because the two cases are worth different things to you.

It was written but did not check out. The archive is complete and wrong. That is worth keeping to look at, so it is renamed out of the way:

`Mp10Data-2026-07-31.failed`

Note the name replaces `.zip` rather than adding to it, so these can never be mistaken for archives or counted as one.

It ran out of room, or could not be written. The file is truncated. There is nothing to learn from half an archive, and keeping a partial copy of a very large backup on a disk that has just filled up only makes the next night worse — so it is deleted, and the failure e-mail says so.

It took longer than `ARCHIVE_MAX_HOURS`. The zip step is stopped, along with everything it started, and the night is reported as a failed archive. Whatever had been written by then is renamed to `.failed` in the same way, so it cannot occupy one of your retention slots as though it were a good backup.

> **A slow archive is not a broken one.** `ARCHIVE_MAX_HOURS` exists only so a zip that will never finish cannot park AutoTasks on it indefinitely — it is not a target. The failure e-mail says how long the archive actually ran; if it was simply still working, raise this number rather than leaving the site with no archive. Zipping a large, image-heavy database can legitimately take many hours.
>
> **Kept `.failed` files are pruned, but only the most recent few survive.** They do not count towards `MAX_BACKUPS_TO_KEEP`; AutoTasks keeps the last three and erases older ones, naming each in the log and in the notice. Treat a `.failed` file as something to investigate, then delete it once you are done.

4.2.3 Restoring from an archive

To get files back out of an archive, use the archiving program directly. From the folder holding the AutoTasks program:

```
python backup_zip.py restore --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --target "D:\Restored"
```

To check an archive without extracting anything:

```
python backup_zip.py verify --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --mode full
```

`verify` finishes with a status of `0` when the archive is sound and `4` when it is not, and writes what it found to `backup_zip.result.json`. The `full` check reads every byte, so on a large archive it takes a while — that is why the nightly run uses a faster structural check instead, and why this thorough one is available to run by hand whenever you want reassurance.

Extracting files is only half of a recovery: what comes out of the zip is a backup *image*, not a database anyone can log into. Turning it back into one is covered in *Restoring the database* at the end of this chapter.

4.2.4 What must be installed for backups to work

What	Why	Where
Python 3	Creates and checks the zip archive	`GENERAL` / `PYTHONPATH` — folder holding `python.exe`, with a trailing backslash. Blank = rely on the system `PATH`
`backup_zip.py`	The archiving program itself	Must sit in the same folder as the AutoTasks program

If either is missing, no archive is created. The database backup itself still runs and your data is still written to `BACKUP_PATH` — but there will be no zip, and you will get a failure e-mail saying so.

4.3 Being told what happened

Backups run at night, when nobody is watching. AutoTasks e-mails a short report each time one finishes — **whether it worked or not** — so you find out without having to go and look.

Entry	Example	Default	What it means
`NOTIFY_EMAIL`	`support@structuredsystems.com`	`support@structuredsystems.com`	Who to notify. Blank = do not send

On a normal night you get **two** messages:

1. when the database backup finishes, and
2. when the zip archive finishes — this one also lists any older archives that were deleted to stay within `MAX_BACKUPS_TO_KEEP`.

The second message is the more useful of the two, because rotation is the only part of the process that destroys anything, and this is the only record you get of it.

```
The Mp10 backup archive was created and verified.

Data dictionary : \\adsserver\data\mp10\mp10.add
Archive         : D:\Backups\Zips\Mp10Data-2026-08-01.zip
Size            : 3072.0 MB (3,221,225,472 bytes)
Files archived  : 428
Source          : \\adsserver\backup\mp10\
Started         : 2026-08-01 04:37:11
Finished        : 2026-08-01 04:52:44
Duration        : 00:15:33
Verification    : headers - passed

Removed         : 2 older archive(s)
                  Mp10Data-2026-06-28.zip
                  Mp10Data-2026-06-29.zip
```

The **Verification** line is the one to read. If it is not there, or the subject says FAILED, nothing was deleted and the archive should not be relied on.

4.3.1 Sending to more than one person

`NOTIFY_EMAIL` takes as many addresses as you like. Separate them with commas or semicolons — you can mix the two, and spaces around them do not matter:

```
support@structuredsystems.com; manager@yourpractice.com, it@yourpractice.com
```

Anything that is not an address is ignored rather than sent, so one typo in the list does not stop the others receiving their copy. Look in `AutoTasks.out` for `Backup notice sent to` to see how many recipients it actually used.

Delivery uses the same `eMail` settings as automated reports (see *Making sure e-mail actually goes out*). If those are not configured, backups still run normally; only the notification is lost, and the log records that it could not be sent.

4.3.2 When something goes wrong

Failures are e-mailed too, with `FAILED` in the subject so you can spot or filter them. There are two, and they mean very different things.

`Mp10 backup FAILED` — the database backup did not happen. There is no new copy of your data today. The message carries the error the database server reported, which is usually enough to identify the cause:

```
The Mp10 hot backup FAILED.  NO BACKUP WAS TAKEN TODAY.

Data dictionary : \\adsserver\data\mp10\mp10.add
Destination     : \\adsserver\backup\mp10\
Started         : 2026-08-01 02:00:03
Failed          : 2026-08-01 02:00:09
Ran for         : 00:00:06

ADS error       : 7077
ADS message     : The specified path is invalid
```

`Mp10 backup archive FAILED` — less serious. The backup itself worked and today's data is safe in `BACKUP_PATH`; only the zip step failed, so there is no dated archive for today and **nothing was rotated or deleted**. Usually a full disk, a permission problem on `BACKUP_ZIP_PATH`, or Python missing from the server.

`Mp10 backup INTERRUPTED` — the backup started and AutoTasks stopped before it finished. Whatever is in `BACKUP_PATH` is half a backup and must not be relied on; earlier archives are untouched. The service was restarted, the server rebooted, or the backup passed `BACKUP_MAX_HOURS` and the hang watchdog restarted it. The log says which: look for a `WATCHDOG` line just before the restart, and if it is there, raise `BACKUP_MAX_HOURS` — the backup was still working when it was cut off.

A backup interrupted **today** is not retried today, for the same reason a failed one is not: an hours-long backup that keeps being cut short would otherwise restart every couple of minutes for the rest of the day. The e-mail says which case you are in.

The message names what actually went wrong and where to look. Two files next to the AutoTasks program carry the detail:

File	What is in it
`backup_zip.log`	What the archiver did, file by file, and why it stopped
`backup_zip.stdout.log`	Anything Python itself printed. **Look here first if `backup_zip.log` does not exist at all** — that combination means Python could not start

> **A failure e-mail means no backup today, and none is coming without you.** AutoTasks does not retry after a failure — see *BACKUP_TIMES* above — so the message is not a warning about a passing glitch. Somebody has to fix the cause and, if the practice cannot go a day without a backup, restart the service.

Silence still means something is wrong: if the service itself is stopped, it cannot send anything at all. A night with no message of any kind is worth looking into.

4.3.3 E-mail has to be set up first

The notices go out through the practice's outgoing mail server, configured in the `eMail` section (see *Making sure e-mail actually goes out*). If that has not been set up, AutoTasks does not attempt to send at all — it writes a line to `AutoTasks.out` naming the settings that are blank and carries on.

Backups themselves are unaffected either way. Notification is the last step, and nothing about it can stop a backup from running or succeeding.

4.4 A working example — backups

Section	Entry	Value
BACKUPS	BACKUP_PATH	\\adsserver\backup\mp10\
BACKUPS	BACKUP_TIMES	02:00:00
BACKUPS	BACKUP_DAYS	234567
BACKUPS	BACKUP_ZIP_PATH	D:\Backups\Zips\
BACKUPS	MAX_BACKUPS_TO_KEEP	30
BACKUPS	NOTIFY_EMAIL	support@structuredsystems.com; manager@yourpractice.com
GENERAL	PYTHONPATH	C:\Program Files\Python313\

This takes a backup Monday through Saturday, any time after 2 a.m., zips it to a local disk, checks the archive, keeps the last 30 zips, and e-mails two people when it is done.

4.5 Backups are not a disaster recovery plan on their own

A backup that has never been restored is a hope, not a backup. Copy the zips off-site — or at minimum onto a different physical machine — and test a restore periodically. AutoTasks writes the file and checks that it is readable; making sure it survives a fire, a theft or a ransomware event is a decision only the practice can make.

Restoring is a single command, and it is worth running once so you know it works before you ever need it:

```
python backup_zip.py restore --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --target "D:\RestoreTest"
```

That command brings the files back out of the zip. Making a working database out of them is the next section.

4.6 Restoring the database

Unzipping an archive is not a restore. The files the nightly backup writes look like a database — tables, memo files, a data dictionary — but Advantage's own documentation is blunt about them: they exist solely to be restored from, and reading them directly produces unpredictable results. Indexes are not in the backup at all. What turns a backup image back into a database people can log into is the **restore**, and it is done from the Admin program.

There are two steps. The first is only needed when the copy you want is inside a zip.

4.6.1 Step 1 — get the backup image onto a disk the server can reach

Last night's backup is already sitting in `BACKUP_PATH`, unzipped. If that is the copy you want, skip straight to step 2.

For an older or off-site copy, extract the archive first, exactly as shown in *Restoring from an archive*:

```
python backup_zip.py restore --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --target "D:\RestoreTest"
```

Where you extract it matters. The restore in step 2 is carried out by the Advantage service **on the server**, so the same rule as `BACKUP_PATH` applies: the extracted folder must be on the server itself, or on a share the server machine can read. A drive letter that exists only on your workstation means nothing to the server.

4.6.2 Step 2 — restore with Admin10

Start Admin10 and log in with user code `adssys`. ADSSYS is Advantage's own administrator account, not an ordinary Mp10 user — its password is the database administrator password, and the restore option does not appear on the menu for anyone else.

> **If your ADSSYS password is still whatever it was on installation day, change it.** Anyone holding it can do everything described here, including overwriting a database.

Then pick **File → Restore DB From Backup**. It asks for three things:

Field	What to enter
Backup image `.add` path	The dictionary file inside the backup image — e.g. `D:\RestoreTest\mp10.add`, or the `.add` directly under `BACKUP_PATH`
Backup dictionary password	The ADSSYS password **as it was when that backup was taken**. If the password has changed since, an old image still wants the old password
Destination `.add` path	Where the restored database should be created — e.g. `D:\RestoredDb\mp10.add`

Both paths are read by the server, so the workstation-drive-letter rule above applies to them too. Both must end in `.add`.

> **The destination must never be the live data directory.** A restore overwrites whatever is at the destination. Restore to a new or scratch directory, look at the result, and only then decide what to do with it — pointing the practice at a restored copy is a deliberate step, best taken with support on the line.

A restore can take a long time on a large database — every table is copied and every index is rebuilt from scratch, which is work the nightly backup skipped. A progress window shows that the restore is alive, how long it has been running and, once the server starts reporting it, how far along it is. Once the server starts reporting progress the restore can be cancelled from that window; a cancelled restore leaves an incomplete copy at the destination, which should be deleted before trying again.

When it finishes you have a complete, working copy of the practice's database at the destination — dictionary, tables and freshly built indexes — completely separate from the live one. That is also the cheapest reassurance there is: restore into a scratch folder once in a while, log into the copy, and you know the night the archive is actually needed, it will work.

4.6.3 When there is no working database to log into

The instructions above assume you can start Admin10 normally. On the day you most need a restore that may not be true: Admin10 connects to a data dictionary at start-up, and the reason you are restoring may be that there is no longer a dictionary to connect to.

There is a way in, and it is worth knowing *before* the bad day.

Point Admin10 at the backup image itself. A backup image contains its own `mp.add`, so it is a data dictionary in its own right and Admin10 can log into it like any other. Start it with the backup folder as the path:

```
PrAdmin10.exe PATH:e:\adsdata\restore\backup\
```

Give the path of the *folder* holding the backup's `mp.add`, with a trailing backslash — Admin10 adds the dictionary name itself. (The same thing can be done by changing the `PATH` entry in `mp10.ini`, but the command line leaves the file alone, which is one less thing to put back afterwards.)

Log in as `adssys` using **the ADSSYS password as it was when that backup was taken**, then use **File → Restore DB From Backup** exactly as described above. You are now connected to the backup, but you still restore *from* it, naming its `.add` as the source and a fresh folder as the destination.

> **Use that session for the restore and nothing else.** Advantage is explicit that backup files exist to be restored from and that reading them directly "can produce unpredictable results" — indexes are not even included in a backup. Admin10 may also tell you the data dictionary version is older than the program expects; that notice is informational and you should ignore it here. Never run a structure update against a backup image — that is your only copy of the data, and it is not the thing to be experimenting on.

If Admin10 cannot open the dictionary at all, it now offers the restore by itself. When the connection fails for any reason other than a rejected password, it asks:

```
The data dictionary could not be opened:
   \\server\share\data\mp.add
   ADS error 7077

Restore a database from a backup image now?
```

Answer yes and you get the restore dialog directly, without logging in — Advantage allows a restore to run on a bare connection to a directory, with no dictionary involved. Answer no and you get the familiar offer to create a new dictionary instead.

Two things about that prompt are deliberate. It does **not** appear when the password was simply wrong, so it is not a way around the login. And once the restore is done Admin10 closes, because it still has no dictionary — start it again pointing at the restored folder, and you are back to a normal session.

4.6.4 The restore utility: MpRestore10

Both routes above still go through Admin10, a program built around having a database. There is also a small separate one that is not:

```
MpRestore10.exe
```

It sits beside the other Mp10 programs and does exactly one thing. There is no login, because there is no dictionary to log into. It asks only for a folder on the server to connect through, and then shows the same restore dialog described earlier, with the same three fields.

The folder it asks for is **not** the database — it is any folder the Advantage service itself can read, used purely to reach the server. Your data folder is the obvious choice and is offered as a starting point (taken from `mp10.ini`), but if that is the part that has gone missing, any other folder on the server does just as well. The usual rule applies: it must be a path the *server* can resolve, so UNC paths and drive letters local to the server work, while a drive mapped on your own PC does not.

> **The Advantage service has to be running.** A restore is carried out by the Advantage Database Server itself, not by the program in front of you. Advantage's own documentation is flat about it: *"Online backup and restore functionality is only supported with the Advantage Database Server."* Advantage Local Server — the mode where there is no service, just files on disk — cannot restore a backup at all, and no Mp10 program can work around that. If the machine holding your data has no Advantage service running, the service is the first thing to fix.

The path can also be given on the command line, handy for a recovery shortcut prepared in advance:

```
MpRestore10.exe PATH:\\server\share\mp10data\
```

> **Prepare this before you need it.** Check once, on a calm afternoon, that MpRestore10 starts and connects — and while you are there, run a restore into a scratch folder with it. A recovery tool nobody has ever opened is the same kind of promise as a backup nobody has ever restored.

5 Automatic error report e-mails

When something goes wrong in a desktop program — PrBill10, Patients10, any of them — Mp10 already asks the person at the keyboard whether to send a report to Structured Systems. What this adds is what happens to that report **whether or not they say yes**: every error a desktop program hits is recorded regardless, and AutoTasks e-mails anything that has not already been sent on its next wake-up (every two minutes by default — see *How AutoTasks works*, above). A user declining to send no longer means the report is lost; it just waits for AutoTasks to pick it up.

5.1 Errors are grouped, not listed one by one

The same error often happens more than once — on the same workstation, or on several. AutoTasks does not send one e-mail per occurrence. It sends **one message per distinct error**, however many times it happened, and the message says how many times and where.

Each message tells you:

- which program and which build,
- which workstations and users hit it, and when it was first and last seen, and
- the list of program locations the error passed through — the stack — so Structured Systems can see where it happened.

It deliberately leaves out the detailed diagnostic dump — the full parameter and variable listing stays in the database, not in your inbox. **It also leaves out machine details:** CPU, memory and the free-resources figures are recorded with the error but never e-mailed. If Structured Systems needs those to chase a resource-exhaustion problem, they come from the database, not from this message.

5.2 Turning it on

It ships switched off. There is no default address — until an administrator sets one, nothing is sent, ever.

Entry	Example	Default	What it means
`NOTIFY_EMAIL`	`support@structuredsystems.com`	*(none — blank)*	Who to notify. Blank = do not send. **This is what switches the feature on**
`MAX_ROWS_TO_KEEP`	`1000`	`1000`	How many error reports to keep before the oldest already-sent ones are deleted. Pruning is part of the error-report pass, so **while `NOTIFY_EMAIL` is blank, or while e-mail itself is not set up, this setting does nothing at all**. Once both are in place it is applied on every wake-up, whether or not there is anything to send — so lowering it takes effect within a couple of minutes on a quiet site, not only after the next error

Entry	Example	Default	What it means
`MAX_GROUPS_PER_RUN`	`10`	`10`	Most distinct errors e-mailed on any one wake-up; anything left over waits for the next one

All three live in section **`ERRORREPORTS`**.

Delivery uses the same mail settings as backups and automated reports — see *E-mail has to be set up first*, above. If those are not configured, AutoTasks does not attempt to send an error report either, and the log records that it could not be sent.

5.3 A worked example

This is reproduced exactly as AutoTasks composes it — labels, spacing and all — because that is what will actually land in your inbox:

```
Subject: [PrBill10.exe] BASE/1005|POSTPAYMENT|2214 - 4x on 2 station(s)

Signature : BASE/1005|POSTPAYMENT|2214
Program   : PrBill10.exe  build 931
Occurred  : 4 time(s)
First seen: 2026-07-30 09:14:02
Last seen : 2026-08-01 11:47:55

Stations and users (oldest to newest)
   FRONTDESK-2         jsmith              2026-07-30 09:14:02   ErrorNum 8841
   FRONTDESK-2         jsmith              2026-07-30 15:02:11   ErrorNum 8852
   BILLING-1           mrivera             2026-08-01 08:30:47   ErrorNum 8901
   BILLING-1           mrivera             2026-08-01 11:47:55   ErrorNum 8919

Stack (most recent report)
------------------------------------------------------------
Called from:POSTPAYMENT(2214)
Called from:APPLYBALANCE(861)
Called from:_BEND(0)
------------------------------------------------------------
Full detail -- including the variables dump and machine/resource stats omitted from this e-mail -- is i
```

The program name in both the subject and the `Program` line is the file name of the running executable, spelled exactly as it sits on disk — `PrBill10.exe`, not `PRBILL10.EXE`. The `build` figure is the internal build number the program was compiled with (a plain counter such as `931`), not the dictionary version.

The signature — `BASE/1005|POSTPAYMENT|2214` above — is an internal fingerprint, not a sentence. It is the part that decides which occurrences belong together: the error subsystem and code (`BASE/1005`), then the routine and the line number inside it. Two occurrences with the same signature are treated as the same error and arrive in one message.

The first line of the message and its closing sentence use internal labels rather than plain English. They are quoted here exactly because that is what will arrive in your inbox — the rest of this guide talks about "the error report" and "the record kept in the database," not those names.

5.4 If sending fails

If the mail server is unreachable or rejects the message, AutoTasks retries on later wake-ups. After three failed attempts it gives up on that report and records the fact in the log — see *The log file*, above.

5.5 What gets deleted, and what never does

Once the number of error reports on file goes over `MAX_ROWS_TO_KEEP`, AutoTasks deletes the oldest ones to bring it back down — but only reports that have already been sent. **A report that has never been sent is never deleted to make room for a new one.** If your site is generating errors faster than they can be e-mailed, the table grows past the keep-limit rather than losing a report nobody has seen yet.

5.6 Where to look when nothing arrives

Everything this feature does — including whether each message actually got out — is written to `AutoTasks.out`, the same log backup notices use. If an error happened and no e-mail followed, that is the first place to look.

6 Electronic remittances: the two kinds

Payment advices reach a practice in two different shapes, and AutoTasks handles them with two separate mechanisms. They are configured independently, in different sections of the System Registry, and turning one on does nothing for the other.

	X12 835 files	**EOB / ERA PDFs**
What it is	The standard electronic remittance format, a data file	A printed advice saved as a PDF
Where it comes from	A clearinghouse, usually downloaded automatically	Downloaded by hand from an insurer's portal
Turned on by	`AUTOTASKS` / `PROCESS_835`	`SCANNING` / `EOBIMAGESPATH`
Folder settings live in	`X12FILE_PATH`	`SCANNING`
Needs Python	No	Yes
Files held back for a person	See that chapter	Moved to a `review` sub-folder, with a reason file

If your clearinghouse delivers 835 files, that is the path to use — it carries more detail and needs no interpretation. The PDF route exists for payers who will only give you a printed advice.

Both post into the same remittances table once processed.

7 Collecting files from the clearinghouse

Rather than have somebody log into the clearinghouse portal and download files by hand, AutoTasks can collect them itself on a schedule. One mechanism brings down everything the clearinghouse has routed to you.

7.1 What gets collected

The clearinghouse tags each file it sends with a document type. AutoTasks saves each one with that type at the front of the file name, so you can tell at a glance what arrived:

Arrives as	What it is	Processed by
`835_….txt`	An electronic remittance — a payment	*X12 835 electronic remittances*
`277_….txt`	A claim status response — what the payer did with a claim	*Claim status responses (277)*
`EDI_….txt`	An acknowledgement or notice about a claim submission	*Claim acknowledgements (EDI)*

> **Everything lands in one folder — the one named by `X12FILE_PATH` / `835` —** whatever its type. That folder is the clearinghouse in-tray, not the "835 folder", despite the setting's name. Each processor then picks out the files belonging to it. This is why the 835 processor only looks at names containing `835`.

A file that has already been collected is not written a second time: AutoTasks checks both the in-tray and the history folder before saving.

7.2 Turning collection on

Section	Entry	Example	What it means
`AUTOTASKS`	`GetRoutedFiles`	`YES`	Collect files from the clearinghouse. Anything other than `YES` = off
`AUTOTASKS`	`GetRoutedFilesStartTime`	`07:00:00`	Earliest time of day it may run. Blank = no lower bound
`AUTOTASKS`	`GetRoutedFilesEndTime`	`19:00:00`	Latest time of day it may run. Blank = no upper bound
`AUTOTASKS`	`GetRoutedFilesFrequency`	`60`	Minutes to wait between runs on the same day
`AUTOTASKS`	`GetRoutedFilesDays`	`23456`	Which days it may run on, using the same day codes as `BACKUP_DAYS`

> **Collection stops at 7 p.m. regardless of what you set.** There is a fixed cut-off in the program, described in the log as *"Past working hrs not fetching routed files to give Inmediata a rest."* Setting `GetRoutedFilesEndTime` later than `19:00:00` will not extend collection past it.

7.3 The clearinghouse account

Section	Entry	What it means
`X12WebService`	`UserName`	Clearinghouse user name
`X12WebService`	`Password`	Clearinghouse password
`X12WebService`	`FileTransfer_Url`	The file-transfer service address
`X12WebService`	`TimeOut`	How long to wait for a response, in milliseconds. Default `30000`
`X12WebService`	`MarkDownloadsAsDownloaded`	`NO` (recommended) — see below

7.4 `MarkDownloadsAsDownloaded` — leave this alone unless asked

Once a file is marked as downloaded, the clearinghouse will not offer it again. When that marking happens is the whole question.

- **`NO` (the default, and the safe choice).** AutoTasks fetches the files, writes them to disk, and *then* tells the clearinghouse they were collected. If the download is interrupted or the disk is full, nothing was marked, so the clearinghouse still has the files and the next run collects them again.
- **`YES`.** The files are marked as downloaded as part of the fetch itself, before AutoTasks has written anything. If the run fails at that point, the clearinghouse considers the files delivered and **will not offer them again** — they would have to be retrieved from the portal by hand.

Set it to `YES` only if the clearinghouse specifically asks you to.

7.5 Checking it is working

```
GetRoutedFiles
7 routed files downloaded
\\server\data\x12files\835_20260803074113.txt
\\server\data\x12files\277_20260803074115.txt
```

A run that finds nothing logs `0 routed files downloaded`, which is normal outside business hours or when the clearinghouse has nothing new.

8 X12 835 electronic remittances

An 835 is the standard electronic remittance file. AutoTasks can collect them from the clearinghouse, read them, post the payments, and file the used ones away — with no manual step at all once it is set up.

8.1 The journey of an 835 file

```
clearinghouse
     |
     |  (1) downloaded  -- AUTOTASKS / GetRoutedFiles
     v
X12FILE_PATH / 835              <- the working folder
     |
     |  (2) processed  -- AUTOTASKS / PROCESS_835
     v
payments posted into Mp10
     |
     |  (3) filed away
     v
X12FILE_PATH / 835_COPY_PATH    <- the history folder
```

The file is **copied** to the history folder and then **erased** from the working folder. Erasing is the point: it is what stops the same remittance being processed again the next time round.

8.2 Step 1 — Create the two folders

You need two folders, and **both must exist before you start**. AutoTasks does not create them for you.

Folder	Purpose
Working folder	Where 835 files arrive and wait to be processed
History folder	Where they are filed after they have been posted

Requirements:

- The AutoTasks service account needs permission to **read, write and delete** in the working folder — it erases files there once they are safely filed.
- It needs permission to **write** in the history folder.
- Keep them separate. Do not make the history folder a subfolder that the working folder's file search would find.
- Do not use the working folder for anything else.

8.3 Step 2 — Declare the folders

Section	Entry	Example value	What it means
`X12FILE_PATH`	`835`	`\\server\data\x12files\`	The working folder — where 835 files are read from
`X12FILE_PATH`	`835_COPY_PATH`	`\\server\data\x12files\history\`	The history folder — where they are moved after posting

Include the trailing backslash on both.

`835_COPY_PATH` **is not optional in practice.** If you leave it blank, processed files are never erased from the working folder, so AutoTasks finds and reprocesses the same remittances on every single cycle, for ever. The same happens if the history folder does not exist or cannot be written to: the copy fails, so the erase is skipped. **If you see the same remittances being processed over and over, this is why.**

8.4 Step 3 — Turn processing on

Section	Entry	Value	What it means
`AUTOTASKS`	`PROCESS_835`	`YES`	Read and post 835 files. Anything other than `YES` = off

8.5 Step 4 — Optional: collect the files automatically

AutoTasks can fetch files from the clearinghouse itself, so nobody has to download anything by hand. That mechanism collects remittances, claim status responses and acknowledgements together — see *Collecting files from the clearinghouse* below.

If you leave `GetRoutedFiles` off, everything else still works; somebody just has to put the 835 files into the working folder themselves.

8.6 Two things that catch people out

The file name has to contain `835`. AutoTasks only looks at files whose name includes those three digits — `*835*`. A remittance saved as `remit_0731.txt` will sit in the folder untouched, for ever, with nothing in the log to say why. If files are being ignored, check the name first.

Files older than six months are skipped. Anything more than 180 days old is left alone and noted in the log as `File more than 6 mo old`. This stops an old archive folder being reposted by accident. If you genuinely need to post something older, it has to be done by hand.

8.7 A working example — 835 files

Section	Entry	Value
`AUTOTASKS`	`PROCESS_835`	`YES`
`X12FILE_PATH`	`835`	`\\server\data\x12files\`
`X12FILE_PATH`	`835_COPY_PATH`	`\\server\data\x12files\history\`
`AUTOTASKS`	`GetRoutedFiles`	`YES`
`AUTOTASKS`	`GetRoutedFilesStartTime`	`07:00:00`
`AUTOTASKS`	`GetRoutedFilesEndTime`	`19:00:00`
`AUTOTASKS`	`GetRoutedFilesFrequency`	`60`
`AUTOTASKS`	`GetRoutedFilesDays`	`23456`

This collects remittances from the clearinghouse once an hour, Monday to Friday, between 7 a.m. and 7 p.m., posts them, and files them into the history folder.

8.8 Checking it is working

At startup AutoTasks writes the state of each X12 folder into `AutoTasks.out`:

```
X12 835 dir  \\server\data\x12files\  exists
```

If that line says the folder is missing, nothing else in this chapter will work. During a run you will see how many files were found:

```
ProcessAll835OnDir  12 Files found on \\server\data\x12files\
```

Then, as each is filed away, the copy and the erase are both logged. A file that is found but never erased is the signature of a history-folder problem — see the warning under *Step 2*.

9 Claim status responses (277)

A 277 tells you what the payer did with a claim you submitted — accepted, rejected, pending, and why. AutoTasks reads them and posts the status against the claim, so the billing staff can see it without logging into a portal.

Section	Entry	Value	What it means
`AUTOTASKS`	`PROCESS_277`	`YES`	Read and post 277 files. Anything other than `YES` = off
`X12FILE_PATH`	`277`	`\\server\data\x12files\`	The folder 277 files are read from
`X12FILE_PATH`	`277_COPY_PATH`	`\\server\data\x12files\history`	Where they are filed after posting

The same rule as for 835 files applies: **the history folder must exist and be writable**, or the file is never erased and gets reprocessed on every cycle.

> **`X12FILE_PATH` / `277` and `X12FILE_PATH` / `835` should normally name the same folder.** Automatic collection puts *everything* it downloads into the folder named by the `835` entry — including 277s. If the `277` entry points somewhere else, downloaded 277 files will sit in the in-tray and never be processed. Unless you have a specific reason to separate them, set both to the same path.

AutoTasks accepts three shapes of status file: a standard X12 277, an XML status report, and a tab-delimited validation report. It works out which it is by reading the file, so you do not have to tell it.

A file that is not any of those is left alone and logged as `Unlegit 277`. That line is normal in a shared in-tray — it is what an 835 looks like to the 277 reader — and is not an error.

10 Claim acknowledgements (EDI)

When you submit claims, the clearinghouse sends back acknowledgements saying it received them. AutoTasks reads these and records them against the claims.

Section	Entry	Value	What it means
AUTOTASKS	ProcessEDIs	YES	Read acknowledgement files. Anything other than YES = off
X12FILE_PATH	277	\\server\data\x12files\	Read from the same folder as 277s
X12FILE_PATH	277_COPY_PATH	\\server\data\x12files\history\	Folder to move afterwards

There are no separate path settings — acknowledgements use the 277 pair.

10.1 TA1 and 999 acknowledgements

Two more files come back for every 837 that is sent, minutes after it is sent, before the claim-level acknowledgement above:

- **TA1** (TA1_….txt) — did the clearinghouse accept the file's envelope.
- **999** (999_….txt) — did each set of claims inside the file pass the 837 syntax checks, and if not, which segment and which element failed.

AutoTasks reads both, works out which claims were in the 837 they answer, and records the result on each claim:

- **Accepted** — a note on the claim (area TA1_ack or 999_ack) saying the file was accepted. The claim's flag does not change: acceptance here only means the file arrived intact.
- **Rejected** — the same note, with the reason (for a 999, the segment, the element and the value that failed), **and** the claim turns red: a rejected entry in its status history and the red flag, unless the claim is already paid, pending or overpaid. The claim then appears on the dashboard's rejected-claims line.

The claims are found by reading the 837 itself, named after the number the acknowledgement quotes (610222951 answers 610222951.837), in the 837 folder or the history folder. If the 837 is no longer there, the claim history notes of recent submissions are searched instead.

Reading the same acknowledgement twice adds nothing. Afterwards the files are filed in the history folder like every other inbound file. An acknowledgement of an eligibility request (a 270), or of an 837 that was not sent from this dictionary, is logged and filed without touching any claim.

11 Claim submissions from Mp10 Web (837)

Billing staff can select claims in Mp10 Web and press **Create 837**. That queues a job; it does not build or send anything. AutoTasks picks the job up on its next sweep, builds the 837 file, and — only after someone presses **Send to clearinghouse** on the web's Submissions page — transmits it on a later sweep. Both halves are deliberate, separate actions, and both happen here, in the service, not in the browser.

Section	Entry	Value	What it means
`AUTOTASKS`	`PROCESS_837_JOBS`	`YES`	Build the 837 files Mp10 Web queues. Anything other than `YES` = off. **The default is `YES`**; set `NO` only at a site that bills from the desktop alone
`X12FILE_PATH`	`837`	`\\server\data\x12files\837\`	Where the built file is written. **The service's account must be able to write here** — a path the desktop can reach but the service cannot fails the job, and the job says so
`X12FILE_PATH`	`835_COPY_PATH`	`\\server\data\x12files\History`	A copy of the file is filed here once the clearinghouse has taken it — the same history folder the 835s use
`X12`	`Copy837_Path`	*(optional)*	Where a built file is copied **only when it was not transmitted**, so it can be handed over by hand. Not a copy of what was sent

Sending needs the clearinghouse web-service account in `sys_registry` under section `X12WebService` — `FileTransfer_Url`, `UserName` and `Password` — the same entries the desktop's claims screen sends with. A site that has never set them gets a plain refusal naming them, and nothing is transmitted; nothing is configured separately for the web.

A job that never leaves *Queued* on the web page means this service is not running against the same dictionary, or `PROCESS_837_JOBS` is not `YES`. The web page cannot see the service and cannot tell you which. To prove the service side from the server without waiting for a sweep:

```
AutoTasks --job=0 --no-send
```

builds every queued job right away and prints what it did, ignoring the switch. **Keep the `--no-send`.** Without it the same command is one full sweep — it builds *and* transmits anything already marked for sending — and the banner it prints says which of the two it is about to do.

Each transmission is recorded against the claims it carried, and the web's **837** button on a claim shows them. Resubmitting a claim that was already sent is allowed on purpose (corrected claims are normal); the web warns, it does not block.

12 Insurance eligibility (270 / 271)

AutoTasks can ask insurers whether a patient's coverage is active — a 270 request out, a 271 response back — and store the answer against the patient, so front-desk staff see current coverage without phoning anyone.

Section	Entry	Default	What it means
`AUTOTASKS`	`AutoSend_270`	`NO`	Send eligibility requests. Anything other than `YES` = off
`X12FILE_PATH`	`270`	`\\server\data\x12files\`	Working folder for eligibility files

> **On an existing installation this is probably already set to `YES`.** Earlier versions turned eligibility on by default, and that value was written into the System Registry the first time the service ran. Changing the default does not change what is already stored. If you do not want eligibility requests going out, check the `AutoSend_270` entry and set it to `NO` yourself.

12.1 When it happens

There is no schedule to configure. Eligibility runs on **every service cycle**, alongside the other tasks, and asks about three groups:

1. **Patients** whose insurance record was added or changed recently
2. **Claims** awaiting submission
3. **Encounters** being prepared

12.2 Why a patient might not be checked

This is the part that generates support calls, because nothing appears in the log to explain a patient who was skipped. A patient's insurance record is only asked about when **all** of these are true:

- the insurance plan is flagged to fetch eligibility — `isFetch271` ticked on the plan record in **Planfile**
- the plan has its **Submitter**, **Receiver** and **GS03** identifiers filled in
- the insurance record was added or changed **within the last 7 days**
- there is a **member/contract number** on the record
- it is the patient's **primary** insurance
- no eligibility response has already come back since the record was last changed

The first two are the usual culprits, and they live on the **plan**, not the patient. If eligibility works for one insurer and not another, compare their plan records before looking anywhere else.

The 7-day window is deliberate: it checks coverage that has recently changed rather than re-asking about every patient in the practice every day. To force a re-check on an older record, open and re-save it — that updates its modified date and brings it back into the window.

13 Appointment reminders by text message

Appointments entered in the calendar can be confirmed by text, so the practice finds out about a cancellation before the slot is wasted rather than after.

AutoTasks watches **every appointments table in the data dictionary** — any table whose name contains `appntmnts` — so a practice running more than one calendar is covered without extra configuration.

13.1 How a reminder happens

```
appointment entered in the calendar
     |
     |  (1) unconfirmed, and inside the alert window
     v
added to the SMS queue (SMSMessages)
     |
     |  (2) sent through Twilio
     v
patient's mobile
     |
     |  (3) patient replies to confirm
     v
appointment marked confirmed
```

Queueing and sending are separate steps, both on every service cycle. A message that cannot be sent stays in the queue and is retried.

13.2 The settings

Section	Entry	Example	Default	What it means
`SMS`	`SEND_APPOINTMENT_REMINDERS`	`YES`	`NO`	The on/off switch
`SMS`	`APPOINTMENT_REMINDERS_ALERT_DAYS`	`7`	`3`	How many days ahead to look. **Must be more than 0 or nothing is sent**
`SMS`	`APPREMINDER-TEXT`	see below	*(blank)*	The message itself
`SMS`	`COOLING_PERIOD`	`1`	`1`	Hours to wait after an appointment is changed before texting about it
`SMS`	`MAX-DAILY-SMS`	`100`	`100`	Most messages to send in one day, as a spend guard
`SMS`	`SMS_STATUS_CALLBACK_URL`	*(a web address)*	*(blank)*	Where Twilio reports delivery status. Blank = no delivery reporting

The Twilio account lives in its own section — **IT task**:

Section	Entry	What it means
TWILIO	ACCOUNT_SID	Twilio account identifier
TWILIO	AUTH_TOKEN	Twilio authentication token
TWILIO	TEL_FROM	The number messages are sent from

13.3 Writing the message

APPREMINDER-TEXT is the text the patient receives. Three placeholders are replaced as each message is built:

Placeholder	Replaced with
[date]	The appointment date and time, e.g. 08/05/2026 2:30 PM
[tel]	The patient's mobile number
[guid]	The appointment's internal reference, for use in a confirmation link

For example:

```
Reminder: your appointment is on [date]. Reply C to confirm or R to reschedule.
```

Keep it short. Long messages are split and billed as more than one.

13.4 Which appointments get a reminder

An appointment is texted when **all** of these are true:

- it is in the future, and within APPOINTMENT_REMINDERS_ALERT_DAYS days
- it has **not** already been confirmed or cancelled
- no reminder has been sent for it already
- there is a mobile number of at least 10 digits on the appointment
- the appointment has a real time on it, not 00:00:00
- it was last changed more than COOLING_PERIOD hours ago

One text per patient per day, however many appointments they have. If a patient is booked twice on the same day, they get a single reminder about the first. This is deliberate — it stops a patient with a morning and an afternoon slot receiving two texts.

13.5 Keeping the cost under control

MAX-DAILY-SMS is a hard stop. Once that many messages have been sent today, the queue is left alone until tomorrow and the log records why:

```
Maximum number of SMS reached.  Total used: 100  Maximum set to: 100
```

If reminders stop partway through a busy day, this is the first thing to check. The messages are not lost — they stay queued and go out the following day, by which time the appointment may have passed.

14 Electronic remittances (EOB / ERA PDFs)

Insurers publish payment advices — remittance notices, ERAs, EOBs — to their provider portals as PDF files. AutoTasks can read those PDFs and post the payments into Mp10 automatically, so nobody has to key them in.

This is the **PDF** route. If your payer sends X12 835 files instead, see the previous chapter — the two are configured separately.

14.1 What you have to do

1. **Choose a folder** where remittance PDFs will be saved.
2. **Declare that folder** in the System Registry.
3. **Save PDFs into it** as you download them.

That is the whole workflow. Everything after step 3 is automatic.

14.2 Step 1 — Choose the folder

Any folder the Mp10 server can reach. A share on your file server is the usual choice, so that billing staff can drop files in from their own workstations.

Requirements:

- The AutoTasks service account needs permission to **read, write and delete** in it. It does not merely read the PDFs — it moves them when it is done.
- If more than one person will save files there, give them all write access.
- Do not point it at a folder used for anything else. AutoTasks processes and then moves *every* PDF it finds there.
- It needs permission to **create sub-folders** there. It makes `review` and `failed` itself the first time it needs them.

14.3 Step 2 — Declare the folder

Section	Entry	Example value
`SCANNING`	`EOBIMAGESPATH`	`\\server\data\x12files\scannedEOBs\`

The screenshot in *Where settings live* shows exactly this entry being edited.

> **Setting this value is the on/off switch.** There is no separate "enable" setting. If `EOBIMAGESPATH` has a value, remittance processing is on; if it is blank, it is off. Include the trailing backslash.

It takes effect on the next cycle — a couple of minutes — with nothing to restart.

14.4 Step 3 — Save your remittance PDFs there

Download the remittance PDF from your insurer's portal as you normally would, and save it into the folder. Nothing else is required — you do not need to rename the file or record it anywhere.

Guidelines:

- **One remittance per PDF.** If the portal offers a combined download of many checks, prefer separate files.
- **Let the download finish before AutoTasks wakes up.** Saving directly into the watched folder over a slow connection can expose a half-written file. If that is a risk, save to your desktop and move the finished file in.

- **Do not put anything else in the folder.** Non-remittance PDFs will be read, found to contain nothing usable, and left behind.

14.5 What happens next

On its next wake-up, AutoTasks first checks that Python can be started, and then, for each PDF in the folder:

1. **Reads the PDF.** It first tries to read the text directly. If the PDF is a scan with no text in it, it falls back to optical character recognition.
2. **Writes a `..._eob.json` file** beside the PDF holding what it found — the check number and date, the payee, and every claim and service line.
3. **Checks the remittance belongs to your practice** (see below).
4. **Matches each line to a claim** and compares the remittance with what is already in the remittances file (see *When a remittance is not imported*).
5. **Posts the payments** into the remittances file, ready to be applied. The rows of one check are written as a whole: either every row goes in, or none does.
6. **Files the PDF and its JSON** according to what happened.

Each file is handled on its own. A damaged or unreadable file is set aside and the rest of the folder carries on.

14.6 What the folders mean

Three sub-folders sit directly inside `EOBIMAGESPATH`:

Folder	What lands there	Who creates it
`history`	A remittance that was imported, and a remittance that is certainly a repeat of one already in Mp10 (see below)	**You** — AutoTasks does not create it
`review`	A remittance AutoTasks would not import by itself. Each file comes with a `<name>.reason.txt` saying why	AutoTasks
`failed`	A PDF the extractor could not read at all, with the JSON it left	AutoTasks

```
...\scannedEOBs\history\C1234567.pdf
...\scannedEOBs\review\C1234567.pdf
...\scannedEOBs\review\C1234567_eob.json
...\scannedEOBs\review\C1234567.reason.txt
...\scannedEOBs\failed\C1234567.pdf
```

> **Create the `history` sub-folder yourself.** Unlike the 835 history folder, this one has no setting — it is always a folder called `history` directly inside `EOBIMAGESPATH`. If it is missing, or the service account cannot write to it, the PDF cannot be moved and the same remittance is read again on every cycle.

A remittance entered from the desktop (the **EOB** tab of *Fetch Claims* in the billing program, then **Save Remit**) is recorded the same way: the rows it loaded from the EOB carry the EOB's file name (rows a check already had keep theirs), and the PDF is moved to `history` when it came from this folder or from `review` (with its JSON and reason file); the billing program creates `history` if it is missing. AutoTasks therefore does not import it again. A PDF picked from elsewhere is left where it is.

`history` does not hold everything that was read. A remittance that was not imported, and is not a certain repeat, is **not** filed there: it waits in `review` so that somebody looks at it.

If you want to know what happened to a particular file, search `AutoTasks.out` for its name.

14.7 When a remittance is not imported

Before anything is written, AutoTasks compares the remittance with what Mp10 already holds, and refuses to import it when it looks like a repeat or when it cannot be sure of the claims. The refusal and its code are logged, and the file goes to `review` (or to `history`, where the table says so).

Code	Meaning	Goes to
`POSTED_CHECK`	The check is already **posted**. Posted checks are never touched	`history`
`SAME_PDF`	The same PDF was imported before, and the earlier import holds rows under this remittance's own claims	`history`
`SAME_PDF_INDIRECT`	The same PDF was imported before, but the earlier import was posted on a different claim of the same visit	`review`
`SAME_DATE_TOTAL`	The remittance has no check number, but another check has the same date, the same total and most of the same claims	`review`
`LINES_ALREADY_PAID`	Most of the lines have already been paid by this payer	`review`
`NO_LINES`	Nothing on the PDF could be read as a payment line	`review`
`NAME_TOO_LONG`	The file name would not fit in the remittance's file-name column	`review`
`DUP_TEST_FAILED`	A database read needed for the repeat check failed, so AutoTasks cannot say it is not a repeat	`review`
`WRITE_FAILED`	A row could not be written, or an error occurred while writing; everything written for that check was rolled back	`review`
`NOTHING_WRITTEN`	Nothing was written: every line was held back (see below), had no usable check number, or was already on the check under another claim	`review`
`JSON_UNREADABLE`	The extractor's file is empty or damaged	`review`
`RUNTIME_ERROR`	An unexpected error stopped the import of that file; the error is in the reason file and the log	`review`

A check that is **in Mp10 but not yet posted** (for instance, an earlier run imported only part of it) is not a refusal. AutoTasks adds only the claims that are missing and leaves the rows already there alone.

A remittance that has no check number is given the next check number in sequence, but only when at least one row is going to be written. A repeat or a remittance that is entirely held back does not use up a number. A check number printed on the PDF that is longer than the check column is cut to the column's width.

14.7.1 Lines that are held back

Some lines are never written unattended. The rest of the check is imported, and each held line is logged as `EOB line skipped:` with the claim key, the key the PDF printed, the amount and the reason:

Reason	Meaning
`UNKNOWN_KEY`	The claim number on the PDF matches no claim
`PAID_OVER_BILLED`	The claim is paid more than it was billed
`PAID_OVER_CHECK`	The claim is paid more than the whole check
`PAID_OVER_CEILING`	One line pays more than the ceiling

The ceiling guards against a misread amount on a scan, where two numbers run together. It defaults to 5000 and is read from the System Registry:

Section	Entry	Default
`SCANNING`	`EOB_PAID_CEILING`	`5000`

Raise it only if your practice really receives single payments above that amount.

14.7.2 Which claim a payment goes to

Where a visit has a primary and a complementary (secondary) claim, a payment goes to the claim the PDF names. A line moves to the complementary claim only when the primary claim was **already paid before this check** and the amount fits what the complementary claim still owes. Otherwise the payment stays on the primary claim. If the primary claim has already received exactly this amount for the same procedure under another check number, the payment is treated as that same check posted again: it stays on the primary and the remittance is refused as `LINES_ALREADY_PAID`.

14.7.3 What staff do with `review`

For each file in `review`, open its `.reason.txt` first. It gives the reason code, the check, where the PDF is (moved here with the JSON, or already in `history` because the extractor archived it earlier), and for each line not written the claim key, the key on the PDF, the procedure code, the amount and the flags. It holds keys and amounts only, no patient names.

- If the code says the remittance is a repeat, check the Remittances window and delete the files.
- If it was held back for a claim number or an amount, enter the remittance by hand: see *Entering a remittance by hand* in the Patients and Billing guide, where the **EOB** tab of *Fetch Claims* can read the PDF for you.
- If you cannot tell, keep the files and ask support.

Nothing retries what is in `review`. When a file has been dealt with, delete it. If a file with the same name arrives there again, the newer one replaces the older.

14.7.4 What staff do with `failed`

`failed` holds a PDF the extractor could not read: it exited with an error or wrote nothing. It is moved there once and is not retried. Look in `python_debug.log` for the reason. If the PDF opens normally, enter it by hand; if Python or a library was missing, fix that and then copy the PDF back into the main folder to try again.

14.8 If your server serves more than one practice

A single AutoTasks server can serve several practices, each with its own data dictionary, all watching the same folder.

AutoTasks reads the **payee** — the provider the cheque was made out to — from each remittance and compares it against the NPIs and provider numbers registered for the practice it is currently connected to.

- If they **match**, the remittance is processed and filed as described above.
- If they **do not match**, the file is **left exactly where it is**, untouched, so the run connected to the practice that owns it can pick it up.

You do not need to sort remittances by practice, and nothing is lost if a file is downloaded into the wrong practice's queue.

> If a remittance carries no identifiable payee at all, it is treated as yours and processed. Most payer layouts do print one; some do not.

14.9 What must be installed on the server

(IT task — but if remittances are not being read, this is the first thing to check.)

Component	Purpose	Registry setting
Python 3	Runs the extractor	`GENERAL` / `PYTHONPATH` — folder holding `python.exe`, with a trailing backslash
`extract_eob.py`	The extractor itself	Must sit in the same folder as the AutoTasks program
Poppler	Renders scanned PDFs to images	Installed at `C:\Mp10\poppler\Library\bin`
Tesseract OCR	Reads text out of scans	Installed at `C:\Program Files\Tesseract-OCR`

Poppler and Tesseract are needed **only for scanned remittances**. If your insurers all publish normal text PDFs, remittances will process without them.

A second log, `python_debug.log`, is written next to the AutoTasks program and holds the extractor's own output. It is the place to look if a specific PDF fails.

AutoTasks checks that Python starts before it touches any PDF in a cycle. If it does not, the log says so, no PDF is moved, and the same files are tried again next cycle. The folder is shared by every practice the server serves, so a missing interpreter must not carry other practices' remittances away.

15 Keeping the CMS reference data current

Several of the claim checks in the Billing module compare a claim against reference data that CMS publishes:

- the **Medicare Physician Fee Schedule** (the modifier-50 checks), updated every quarter,
- the **NCCI tables** (code pairs and unit limits), updated every quarter, and
- the **ICD-10-CM code set** (every diagnosis check), updated every October 1st.

AutoTasks keeps all three current on its own. Once a day it looks at the CMS website; when a new quarter or a new ICD-10 year has been published, it downloads the files and imports them into your data. Nothing needs to be remembered, and the import menus in Admin10 remain available for loading a file by hand.

15.1 Turning it off

This is on by default. To switch it off entirely, set:

Section	Entry	Value
AUTOTASKS	FETCH_CMS	NO

A dictionary that has never had the fee schedule or NCCI tables loaded is skipped automatically for those two — the related claim checks then simply place no tags — so leaving FETCH_CMS on is harmless everywhere.

15.2 The settings

Section	Entry	Default	Meaning
AUTOTASKS	FETCH_CMS	YES	The whole feature
AUTOTASKS	CMSFETCH_CHECK_HOURS	24	Hours between looks at the CMS site
AUTOTASKS	CMSFETCH_PATH	empty	Where downloads land. Empty means a `cms` folder beside AutoTasks.exe, one subfolder per quarter plus one per ICD-10 year
CMSFETCH	NOTIFY_EMAIL	empty	Who is e-mailed what a check did; commas between addresses

AutoTasks also keeps its own bookkeeping under the **CMSFETCH** section: `LAST_CHECK`, `ICD10_YEAR` (the latest ICD-10 year downloaded) and `ICD10_IMPORTED` (the latest year imported). Those entries are written by the service; leave them alone unless this guide says otherwise.

15.3 Being told what happened

After any check that imported, downloaded or failed something, AutoTasks sends one e-mail listing each event — for example that the October fee schedule was imported, or that the new ICD-10-CM year was downloaded and applied. A quiet check, when CMS has published nothing new, sends nothing.

The recipients come from **CMSFETCH / NOTIFY_EMAIL**. If that is empty, the error-report address (**ERRORREPORTS / NOTIFY_EMAIL**) is used instead; if both are empty, no mail is sent and the log still records everything. As with every notice AutoTasks sends, e-mail must be set up first — see **E-mail has to be set up first** in the backups chapter.

15.4 The yearly ICD-10 release

The ICD-10-CM set changes every October 1st. AutoTasks downloads the year's order file and addendum, imports them — new codes added, retired codes end-dated so older claims still read correctly, titles and laterality updated — and writes a **report of every change** beside the downloaded files. The e-mail names that report.

If the import cannot run — most often because another session had the diagnosis table open at that moment — nothing is changed, the e-mail says so, and AutoTasks tries again on its next check until it succeeds.

A database that has not had its Data Dictionary updated yet cannot take the import: its diagnosis table lacks the columns the import needs. There, AutoTasks downloads the files, says so in one e-mail, and then waits quietly. The first check after the Data Dictionary update imports the year on its own; nothing has to be run by hand.

To load a year by hand instead (or on a machine where AutoTasks is not running), use **Admin10 → Add 20xx ICD10 Data** against the downloaded `icd10cm_order_YYYY.txt`, then set **CMSFETCH / ICD10_IMPORTED** to that year so AutoTasks does not import it again. Clearing ICD10_IMPORTED makes AutoTasks import that year again at its next check.

15.5 The Index to Diseases (searching diagnoses by words)

The Billing module's ICD10 tab finds codes from plain words ("left kidney stone"). Besides the code descriptions, it searches the CMS **Index to Diseases and Injuries** and the Neoplasm Table. AutoTasks downloads them with the year's code files, into the same `icd10-YYYY` folder, and loads them into your data on its own. **CMSFETCH / ICD10_INDEX_IMPORTED** records the year loaded.

Until the Data Dictionary has been updated, the table the index goes into does not exist. AutoTasks then waits quietly and loads the index at the first check after the update. The search still works meanwhile, from the code descriptions alone.

To load it sooner, use **Admin10 → Load ICD10 Index** and pick the folder that holds `icd10cm_index_YYYY.xml`.

15.6 Diagnosis suggestions by procedure

The ICD10 tab also suggests the diagnoses most often billed with the claim's procedure codes. AutoTasks recounts those every night, at its first check after 21:00, from the primary claims of the last 24 months. On a large practice this takes under a minute.

Section	Entry	Default	Meaning
BILLING	ICD10-SUGGEST-MONTHS	24	How many months back a claim still counts
BILLING	ICD10-SUGGEST-BUILT	—	The day of the last recount, written by AutoTasks

If a night's recount fails, it is tried again an hour later. Clearing ICD10-SUGGEST-BUILT makes AutoTasks recount at its next check after 21:00. **Admin10 → Rebuild ICD10 Suggestions** recounts right away, for example on the day the Data Dictionary is updated.

15.7 Checking it is working

- The log (`AutoTasks.out`) carries one `CMSFETCH:` line per decision: what was current, what was imported, what is not posted yet. "Not posted yet" close to a quarter change is normal and can last weeks — CMS publishes on its own schedule, and no e-mail is sent for it.
- The downloaded files are under the CMSFETCH_PATH folder.

- In the Billing module, the dashboard's claim checks that depend on this data (the modifier-50, NCCI and MUE tags) only work where the data has been loaded.

16 Automated reports

Every Billing and Patients report with an **Automate Report** tab can run by themselves on a schedule and be e-mailed out. You set this up from the report itself, not from a central screen. AutoTasks does the running, so it must be installed and running on the server. If it isn't, nothing is sent.

16.1 Before you start

- **Who can set one up:** only members of the **Supervisors** or **Administrators** groups. Other users can see the **Automate Report** tab but cannot save from it.
- **Who can receive one:** only members of the **BillingUsers** group appear in the recipient list. That is true in the Patients module too. If someone is missing, add them to that group in the admin module, or ask your Mp10 administrator to.
- **Outgoing e-mail must work.** See *Making sure e-mail actually goes out* below. Without it, reports run and nobody receives them.

16.2 Setting up a report

Open the report as you normally would. Set its parameters on the other tabs (insurances, doctors, encounter types and so on), then go to the **Automate Report** tab. The filters you chose on the other tabs are saved with the automation and used every time it runs.

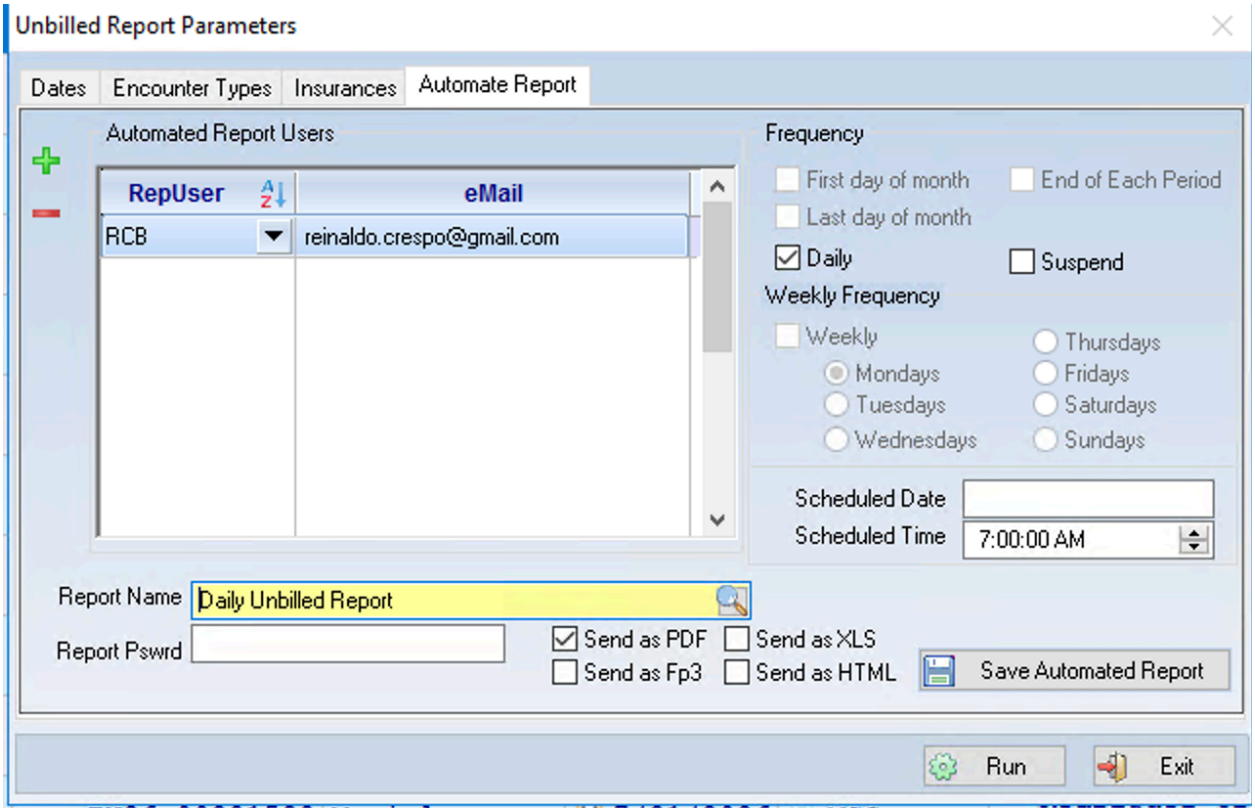


Figure 2: The Automate Report tab, here on the Unbilled Report

16.2.1 Who receives it

In **Automated Report Users**, press the green + and pick a user from the **RepUser** drop-down. Their e-mail address fills in beside them. Add as many recipients as you need; the red − removes the selected one.

The address comes from the user's Mp10 record. If it is wrong or missing, fix it on the user and pick them again here. A recipient with no e-mail address is dropped when you save.

16.2.2 When it runs

Tick one of the **Frequency** options. The option also decides which dates the report covers, because an automated report ignores the dates on the **Dates** tab:

Option	Runs	Covers
Daily	Every day	Yesterday
Weekly	Once a week, on the day selected below it	The seven days before the run day, ending yesterday
First day of month	On the 1st	The whole previous month
Last day of month	On the last day of the month	The current month
End of Each Period	On your billing period's closing day	The whole previous calendar month
Scheduled Date	Once, on that date	The dates you set on the **Dates** tab
Suspend	Not at all (see *Stopping a report*)	—

End of Each Period uses the day of the month set in `BILLING` / `DAYS_INTO_MONTH_BEFORE_CLOSING`. If that setting is `0` or missing, these reports never run.

Scheduled Date is the only way to send a fixed date range, such as a one-off report for last quarter. Set the range on the **Dates** tab first, then pick the date it should go out.

Ticking **Daily** clears the monthly and weekly options, because a daily report cannot also be monthly. Choose **Daily**, or any combination of the others.

16.2.3 What time it runs

Scheduled Time sets the time of day. A new automation starts with the system default:

Setting	Section	Default
`AUTO_REPORTS_TIME`	`AUTOTASKS`	`22:00:00`

> **A report is only queued for a time that has not yet arrived today.** If it is 3 p.m. and you schedule a daily report for 2 p.m., it will not run this afternoon — it starts tomorrow. When setting up a report you want to see today, pick a time that is still ahead of you.

The report goes out on AutoTasks' first wake-up after that time, so allow a couple of minutes (see *How AutoTasks works*).

16.2.4 What format it arrives in

Tick one or more of the following. Each format arrives as a **separate e-mail**, so ticking PDF and XLS sends every recipient two messages.

Format	Good for	Password protected?
Send as PDF	Reading and printing; the usual choice	Yes
Send as XLS	Recipients who want to sort, filter or total the figures	Yes

Format	Good for	Password protected?
Send as Fp3	Opening in the FastReport report viewer	No
Send as HTML	Reading on a phone or in a browser without opening an attachment viewer	No

If you tick none, the report is sent as PDF.

An HTML report arrives as an `.html` file. Any logos or charts in the report come as separate picture files attached to the same e-mail.

Report Pswrd protects the PDF and XLS attachments: the recipient needs it to open them. Use it for anything leaving the practice, and give recipients the password some other way, never in the same e-mail. It does not protect Fp3 or HTML attachments, so don't use those formats for patient information sent outside the practice.

16.2.5 Naming and saving

Give it a clear **Report Name**. This name identifies the automation, and it appears in the subject of every e-mail. For example, a PDF arrives as *"PDF formatted Daily Unbilled Report Automated Report"*. `Daily Unbilled Report` is a better name than `report1`.

Press **Save Automated Report**.

Run does not send anything. It shows the report on screen with the current parameters, the same as running it by hand. Use it to check the filters return what you expect. To see the e-mail itself, schedule the report a few minutes ahead and wait for it.

16.2.6 More than one schedule for the same report

You can save as many automations of one report as you like, each with its own name, recipients, filters and timing. For example, keep a *Daily Unbilled Report* for the billing staff and a *Monthly Unbilled Report* for management.

16.3 Changing an existing automated report

1. Open the same report and go to the **Automate Report** tab.
2. Press the magnifying glass in **Report Name**. It lists the automations already saved for this report. Pick one, and its recipients, timing, formats and filters load into the screen.
3. Make your changes and press **Save Automated Report**.

   Do not rename it while changing it. The name is how the automation is identified. Saving under a new name creates a *second* automation; the old one keeps running under the old name. To rename, save under the new name and then suspend the old one.

16.4 Stopping a report

Tick **Suspend** and save. The automation keeps its recipients and settings but stops running. Untick it and save to resume.

Mp10 has no button for deleting an automation, so suspending is how you retire one.

16.5 Making sure e-mail actually goes out

Scheduled reports are e-mailed through the practice's outgoing mail server. If that is not configured, reports will run and produce nothing anybody sees.

These live in section **eMail** *(IT task)*:

Entry	Example	What it is
SMTP-ServerName	smtp.gmail.com	Outgoing mail server
SMTP-Port	465	Its port
SMTP-User	reports@yourpractice.com	Account used to send
SMTP-Pass	*(password)*	Its password
Address	reports@yourpractice.com	Address messages come from
SMTP-Auth	YES	Whether the server requires a login
SENDMETHOD	CDO or SMTP	How mail is sent

> **If you use Gmail or Microsoft 365**, an ordinary account password will not work. You need an app-specific password, and the account must permit it. This is the most common reason scheduled reports stop arriving after an otherwise unrelated IT change.

16.6 Keeping a copy on disk

To have every automated report also written to a folder (useful as an audit trail, or to feed a shared drive):

Section	Entry	Example
AUTOREPORTS	COPY_TO_PATH	\\fileserver\mp10\reports\

Leave it blank to skip. The folder must exist, and the account AutoTasks runs under must be able to write to it.

16.7 When a report does not arrive

Open the AutoTasks log (see *The log file*) and search for the report's name. Each step leaves a line:

What the log says	What it means
AutoRep: <name> … queued	Scheduled for today and waiting for its time
not queued: not scheduled for today	The frequency doesn't fall on today. Check the ticked options and the weekday
not queued: suspended	**Suspend** is ticked
not queued: run time already past	Today's run time has passed. Either it already ran today (look for its Report e-mail line), or it was saved after its time and starts on the next scheduled day
already on queue	Waiting for its time; nothing wrong
Report e-mail SENT … to: …	Delivered to the mail server, with the addresses it went to
Report e-mail FAILED	The mail server refused it. Check the eMail settings
Report NOT e-mailed -- no recipients configured	The report ran but the recipient list was empty
AutoRep: not run -- AutoTasks cannot build this report	The AutoTasks installed is older than the Mp10 module the automation was saved from. Update AutoTasks

AutoTasks writes one `AutoRep:` line for every saved automation on every wake-up. If the report's name never appears, either AutoTasks is not running or it is connected to a different database than the one where the report was saved.

17 Modality worklist (imaging)

If your practice has imaging equipment — a CT, an ultrasound, an X-ray room — the **modality worklist** is how each machine finds out who its next patient is. Instead of a technologist typing the name, date of birth and accession number at the console, the machine asks Mp10 and fills its own screen in.

That matters for more than convenience. A study labelled by hand is a study that can be labelled wrongly, and a mistyped accession number is what makes an image arrive in the wrong patient's file.

> **This chapter covers the AutoTasks half only.** AutoTasks decides which studies belong on the worklist and writes one small file per study. A **DICOM worklist server** — a separate piece, installed once on your server — is what the machines actually query, and it serves those files. Without it, AutoTasks is writing files nothing reads.
>
> If you are setting this up for the first time, read the **Mp10 Modality Worklist** guide instead: it covers both halves, installing the server, pointing a scanner at it, and what to do when the worklist looks empty. This chapter is the AutoTasks settings reference that guide refers back to.

17.1 Turning it on — two separate switches

There are two, in this order, and the order matters:

```
Admin10 -> Global System Registry
     |
     |  (1) GENERAL / MODALITY-WORKLIST = YES
     v
run a file check in Admin10
     |                      -> the modalitywl table is created
     v
     |  (2) AUTOTASKS / EXPORT_MWL = YES
     v
restart AutoTasks
                            -> the table is filled, and kept current
                               on every cycle from then on
```

17.1.1 Why the settings are already there, set to `NO`

The first time Mp10 looks for a setting and does not find it, **it creates the entry itself**, set to its default, with a description of what it does. So the first time you go looking for `MODALITY-WORKLIST` you will probably find it already present and set to `NO`.

That is not a fault and not somebody else's change — it is Mp10 writing down a question it just asked. Once the entry exists, **your value always wins**; Mp10 never overwrites a value you have set.

17.1.2 Step 1 — create the table

Section	Entry	Set to
`GENERAL`	`MODALITY-WORKLIST`	`YES`

Then **run a file check in Admin10**. The table is created during that check and at no other time.

> **Order matters.** The file check reads this setting as it runs. Setting it to `YES` *after* a check has finished does nothing until the **next** check. If you have just changed it, run the check again.

17.1.3 Step 2 — turn the reconcile on

Section	Entry	Set to
AUTOTASKS	EXPORT_MWL	YES

That takes effect on the next cycle, a couple of minutes later; settings are re-read every cycle and nothing needs restarting.

17.2 The settings

Section	Entry	Example	Default	What it means
GENERAL	MODALITY-WORKLIST	YES	NO	Whether the modalitywl table exists at all. Read during a file check, not by AutoTasks
AUTOTASKS	EXPORT_MWL	YES	NO	The on/off switch for keeping the table current
AUTOTASKS	MWL_SPOOL_PATH	C:\Mp10Mwl\worklists	*(blank)*	Where the worklist files are written. **Must match the worklist server's own setting exactly.** Blank means no files are written
AUTOTASKS	MWL_WINDOW_DAYS	1	1	How far either side of today to look, in days
AUTOTASKS	MWL_UID_ROOT	*(blank)*	*(blank)*	**IT task.** Leave blank unless your organisation owns a registered DICOM root

17.2.1 MODALITY-WORKLIST — whether the table exists

Leave this NO and the table is never created, nothing else in this chapter applies, and no harm is done. Practices with no imaging equipment should leave it alone.

Setting it back to NO later does **not** delete the table or its contents. It only stops future file checks from maintaining it.

17.2.2 EXPORT_MWL — whether the table is kept current

With this NO, the table sits exactly as it was last left. With it YES, AutoTasks re-checks the whole worklist against your schedule on every cycle — by default every two minutes.

17.2.3 `MWL_SPOOL_PATH` — where the worklist files go

The worklist server watches one directory. This tells AutoTasks to write into that same directory, and the two settings must name **exactly the same place**.

Leave it blank and the table is still kept current but no files are written — which is a reasonable way to run for a while before the imaging side exists, and is what the log line `no .wl written -- AUTOTASKS/MWL_SPOOL_PATH is empty` is telling you.

> **Wrong path, empty worklist, no error.** If the two settings disagree, each half does exactly what it was told: AutoTasks writes files, the server serves the empty directory it was pointed at, and nothing anywhere reports a problem. The scanner simply shows nothing. `Test-Mwl.ps1` on the server compares the two in one line, and is the fastest way to rule this out.

AutoTasks owns that directory. It writes what belongs there and deletes everything else in it — that is how a cancelled study disappears from the machines. Do not keep anything else in it.

17.2.4 `MWL_WINDOW_DAYS` — how wide the window is

`1` means yesterday, today and tomorrow. `0` means today only. `7` means a fortnight centred on today.

> **Widen this with care.** Everything inside the window is a live worklist entry that a technologist has to look past to find today's patient. A window of `30` is not a longer list so much as a worse one. Change it only if orders are genuinely being scheduled that far ahead.

Narrowing it does not lose anything permanently — see *Cancelling is not deleting* below.

17.2.5 `MWL_UID_ROOT` — how studies are identified *(IT task)*

Every study needs an identifier that is unique **worldwide**, not just within your practice, because images may end up in systems shared with other organisations.

Left blank — the default, and the right answer for almost everyone — Mp10 mints one from a scheme designed for exactly this and requiring no registration.

Fill it in **only** if your organisation genuinely owns a registered DICOM root. Digits and dots only.

> **Do not invent a value here.** An invented root is not merely untidy: it can collide with identifiers issued by whoever really owns it, and two different studies sharing an identifier is a problem that surfaces years later, in somebody else's archive.

Changing this later is safe. Identifiers already issued are **never rewritten** — only newly-scheduled studies use the new root — so an existing study is never split in two.

17.3 Which orders appear on the worklist

An order is on the worklist when **all** of these are true:

- its encounter is **not** inactive,
- its encounter is **not** closed, and
- its start date falls within `MWL_WINDOW_DAYS` days either side of today.

Where an order has no start date of its own, the encounter's admission date is used instead.

There is nothing to switch on per order and nothing to tick. An order that meets those three tests is on the worklist; one that stops meeting them comes off it.

17.4 What happens on each cycle

AutoTasks works out the list from scratch every time, rather than trying to remember what changed. That is deliberate: it means a missed cycle, a service restart or a crash cannot leave the worklist permanently out of step. The next cycle simply puts it right.

Each pass does four things:

1. **Works out which orders belong** on the worklist right now.
2. **Adds** the ones that are not there yet, and **puts back** any that had been cancelled but now qualify again.
3. **Cancels** anything on the worklist that no longer qualifies.
4. **Writes a file for every study a machine should see**, and **deletes every other file** in the worklist directory. That last sweep is what actually withdraws a cancelled study from the machines.

Step 4 rewrites a file only when its content has changed — or when the file has gone missing, which is how a directory somebody has cleaned out repairs itself on the next cycle.

17.5 Cancelling is not deleting

When an order falls off the worklist — the encounter is closed, or the date moves out of the window — its entry is **marked cancelled, not removed**.

This matters if it comes back. Re-open the encounter, or move the date back into the window, and the *same* entry is reinstated with the *same* study identifier. The study is never split into two half-studies in your image archive.

The one case that does remove an entry outright is deleting the order itself, which is as it should be — an order that no longer exists cannot be scanned.

17.6 Telling the machine what kind of study it is

A worklist entry has to say whether the study is a CT, an ultrasound, an X-ray and so on. Mp10 works that out from the **revenue code** on the order: each revenue code carries the kind of imaging it represents.

Until those are filled in, entries are still created correctly in every other respect — they simply do not say what kind of study they are, which is not enough for a machine to act on. **This is a one-off data job**, done once per practice, and AutoTasks tells you how much of it is left (below).

17.7 What the log tells you

Every cycle writes a handful of lines to `AutoTasks.out`. Search it for `WriteWorklistFiles`:

Line	What it means
`active set size`	How many orders qualify for the worklist right now
`inserted` / `re-activated` / `unchanged`	What this cycle changed. On a quiet cycle all three are `0` and that is correct
`cancelled`	How many entries came off the worklist this cycle
`rows with a rev_code but no revcodes.Modality mapped`	How many entries cannot say what kind of study they are — with example revenue codes, so you know which ones to fill in. **This number should fall to zero** as the mapping is completed
`rows with no items/RevCode match at all`	Orders whose item does not resolve to a revenue code at all. Usually a catalogue problem rather than a worklist one

Then search it for `EmitWorklistFiles` — that is the part the machines see:

Line	What it means
`written`	Files written this cycle. High on the first run of the day, then usually `0`
`unchanged`	Studies already published and unaltered. This is the number that should match what a scanner sees
`purged`	Files removed — cancelled studies, studies past the window, and anything else that was in the directory
`no modality (not emitted)`	Studies held back because their revenue code does not say what kind of study they are. **These are invisible to every machine.** Non-imaging orders — laboratory, office visits — are counted here too and belong here
`write failures`	Should always be `0`. Anything else is a permissions or disk problem on the worklist directory
`published with NO station`	How many studies carry no imaging station. They answer a modality-and-date query but are invisible to a scanner filtering by station AE title. The line names example location/modality pairs — those are the missing **Imaging Stations** rows
`station routing is OFF`	The imaging stations table is absent or unreadable. The worklist is still published; nothing is routed
`no .wl written -- AUTOTASKS/MWL_SPOOL_PATH is empty`	The path is not set, so nothing is published
`ABORTED -- AUTOTASKS/MWL_SPOOL_PATH does not exist`	The path is set but wrong, or the directory was deleted

A healthy log on a normal day shows a steady `active set size`, small or zero change counts, a zero mapping count, and `written 0 / unchanged <today's studies>` on most cycles.

17.8 Running one cycle by hand

You do not have to wait two minutes and read around everything else the service does. On the server:

```
AutoTasks.exe --mwl-run
```

That runs **one worklist cycle and nothing else** — no backup, no remittances, no e-mail, no texts — and prints the same lines to the screen. It is safe to run in the middle of a working day, and it is the quickest answer to "is the worklist right *now*".

18 Troubleshooting

Start with `AutoTasks.out`. Search it for the date you expected something to happen. Almost every question below is answered by a line in that file.

18.1 Nothing at all is happening

Check	How
Is the service running?	Windows **Services**, look for the Mp10 AutoTasks service *(IT task)*
Is it writing to the log?	Check the timestamp on `AutoTasks.out` — it should update every couple of minutes
Were settings changed less than a cycle ago?	Wait one cycle — they are re-read every time, no restart needed. If a change has still not taken effect after that, check the spelling of the entry: a misspelt one is a setting nobody reads

18.2 The backup never runs

Symptom	Likely cause
Nothing in the log about a backup	One of `BACKUP_PATH`, `BACKUP_TIMES`, `BACKUP_DAYS` is blank — all three are required
"Backup started" then an error	The path is not reachable *by the database server*. Check for a mapped drive letter; use UNC instead
Runs some days but not others	Check the digits in `BACKUP_DAYS` against the day-code table — 1 is Sunday, not Monday
Ran once, never again the same day	Working as designed — one backup per calendar day
Zips are disappearing	`MAX_BACKUPS_TO_KEEP` is pruning them. Raise it or move zips off elsewhere

18.3 No backup e-mail arrived

Check the log first — it distinguishes "the backup did not run" from "it ran but the e-mail did not go out", and those need different fixes.

In `AutoTasks.out`	Meaning
`Backup notice sent to N recipient(s)`	It was sent. Look in the recipients' spam folder
`Backup notice FAILED to send to`	The mail server rejected it — check the `eMail` settings
`Backup notice could not be sent (mail error)`	The mail server could not be reached at all
`notifications are switched off`	`NOTIFY_EMAIL` is blank
`holds no usable address`	Nothing in `NOTIFY_EMAIL` looks like an address — check for a typo
Nothing at all about a notice	The backup itself never completed. See *The backup never runs*

Symptom	Likely cause
Only one message instead of two	The second comes from the zip step. If `BACKUP_ZIP_PATH` is blank there is nothing to archive, and the first message says so
Some recipients got it, others did not	An entry in the list is not a valid address and was skipped. The log line gives the count actually used
A `FAILED` message, then nothing more that day	Working as designed — AutoTasks does not retry a failed backup until the next day
No message at all, on a night you expected one	The service is not running, or never reached the backup. Silence is not confirmation — check the log
`No outgoing mail server is configured` in the log	The `eMail` settings are blank. The backup still ran; only the notice was skipped
`already sent today; not repeating` in the log	A notice of that kind had already gone out today

18.4 Remittance PDFs are not being picked up

Symptom	Likely cause
Files sit in the folder, nothing in the log	`SCANNING` / `EOBIMAGESPATH` is blank or misspelled, or the service was not restarted
"belongs to another practice - left in place"	Working as designed — the cheque is payable to a different practice on this server
The file was read but no payments appeared	The claims on it could not be matched. The log names each claim it could not resolve
Log says `EOB Python interpreter could not be started`	Python is not installed, or `GENERAL` / `PYTHONPATH` is wrong. Nothing was moved; fix it and the files are picked up next cycle
Log says `EOB Python script not found`	`extract_eob.py` is not next to the AutoTasks program
A PDF is in the `failed` sub-folder	The extractor could not read it. The log line reads `extract_eob.py failed, code:`; look in `python_debug.log`. Enter the remittance by hand, or fix the cause and copy the PDF back
Only scanned remittances fail	Poppler or Tesseract is not installed
A PDF is in the `review` sub-folder	AutoTasks would not import it by itself. Open the `.reason.txt` beside it: the first lines give the reason code (see *When a remittance is not imported*). Enter the remittance by hand if it is not a repeat
Log says `EOB refused:` followed by a code	The same reason code as in the `.reason.txt`. `POSTED_CHECK` and a direct `SAME_PDF` are certain repeats and go to `history`; the others go to `review`
Log says `EOB line skipped:`	One line was held back (`UNKNOWN_KEY`, `PAID_OVER_BILLED`, `PAID_OVER_CHECK` or `PAID_OVER_CEILING`); the rest of the check was imported. If a genuine large payment is held, raise `SCANNING` / `EOB_PAID_CEILING`
A `..._eob.json` stays in the main folder, log says `TX_ALREADY_OPEN`	Another operation had a database transaction open on the AutoTasks connection, so nothing was read or written. AutoTasks closes it after the remittance pass (next row) and the file is retried on the next cycle

Symptom	Likely cause
`A transaction left open by ... was rolled back` in the log	A job ended with its database transaction still open, and AutoTasks rolled it back so later jobs do not write into it. Once is harmless; every cycle means a defect — send the log to support
`... could not be rolled back: disconnecting` in the log	The rollback failed too (usually the connection to the server is broken). AutoTasks dropped the connection, which makes the server roll the transaction back, and connected again. The remittances not yet read wait for the next cycle
`EOB JSON stopped by a runtime error` in the log	That one file went to `review` as `RUNTIME_ERROR` and the cycle carried on. Send the log line to support

Once processed, an imported PDF is in the `history` sub-folder — it has not been lost. A PDF AutoTasks would not import is in `review` or `failed`. AutoTasks creates the `history`, `review` and `failed` sub-folders when they are missing; it needs permission to create folders in the remittance folder.

18.5 835 files are not being processed

Symptom	Likely cause
The same remittances are processed over and over	`X12FILE_PATH` / `835_COPY_PATH` is blank, or the history folder does not exist, or the service cannot write to it. The copy fails, so the erase is skipped. This is the most common 835 problem
Files sit in the folder, nothing in the log	`AUTOTASKS` / `PROCESS_835` is not `YES`, or the service was not restarted after changing it
Some files are processed, others ignored entirely	The ignored ones do not have `835` in the file name. Only `*835*` is looked at
`File more than 6 mo old` in the log	Working as designed — anything over 180 days is skipped
`X12 835 dir ... does not exist` at startup	`X12FILE_PATH` / `835` is wrong, or is a Windows path on a Linux server
`is not a legit 835` in the log	The file is not a valid 835 — often an acknowledgement or a report the clearinghouse routed alongside the remittances
Nothing is ever downloaded	`AUTOTASKS` / `GetRoutedFiles` is not `YES`, today is not in `GetRoutedFilesDays`, or the time is outside the start/end window
`Error connecting to Host` in the log	The `X12WebService` user name or password is wrong, or the server cannot reach the clearinghouse

18.6 Nothing is being downloaded from the clearinghouse

Symptom	Likely cause
`0 routed files downloaded`, every time	Normal if the clearinghouse has nothing new. If it never changes, check `AUTOTASKS` / `GetRoutedFiles` is `YES`
Nothing after 7 p.m.	Working as designed — there is a fixed cut-off no setting can move

Symptom	Likely cause
Nothing on certain days	Today is not in `GetRoutedFilesDays`
`Error connecting to Host`	`X12WebService` user name or password, or the server cannot reach the clearinghouse
Files download but 277s are never processed	`X12FILE_PATH` / `277` points somewhere other than `X12FILE_PATH` / `835`. Everything is downloaded into the `835` folder
A file was collected once and never again	It was marked as downloaded at the clearinghouse. Retrieve it from the portal by hand

18.7 Claim statuses or acknowledgements are not appearing

Symptom	Likely cause
Nothing happens at all	`AUTOTASKS` / `PROCESS_277` or `ProcessEDIs` is not `YES`
The same files process repeatedly	`X12FILE_PATH` / `277_COPY_PATH` is blank, missing, or not writable
`Unlegit 277` in the log	Normal — that is a file in the shared in-tray that is not a 277, most often an 835
A claim was sent but nothing came back	Look at the claim's notes for a `TA1_ack` / `999_ack` note. None at all: the clearinghouse has not answered yet, or `277` processing is off. A rejected 999 names the segment and element to fix

18.8 Eligibility is not being checked

Symptom	Likely cause
No patients at all are checked	`AUTOTASKS` / `AutoSend_270` has been set to something other than `YES`
One insurer works, another does not	The failing insurer's plan record: `isFetch271` not ticked, or Submitter / Receiver / GS03 blank
An established patient is never checked	Their insurance record has not changed in the last 7 days. Re-save it to bring it back into the window
A secondary insurance is ignored	Only the primary is checked

18.9 The modality worklist is empty or wrong

Symptom	Likely cause
The `modalitywl` table was never created	`GENERAL` / `MODALITY-WORKLIST` was set to `YES` *after* the last file check, not before. Run the file check again
The table exists but stays empty	`AUTOTASKS` / `EXPORT_MWL` is not `YES`, or the service was not restarted after it was set
`active set size` is `0` but there are orders today	Their encounters are closed or inactive — both are excluded — or the order dates fall outside `MWL_WINDOW_DAYS`

Symptom	Likely cause
Everything is suddenly cancelled	`MWL_WINDOW_DAYS` was narrowed, or a batch of encounters was closed. Nothing is lost; widen it or re-open them and the same entries come back
Entries exist but say nothing about the kind of study	The revenue-code mapping is not filled in. The log line counting these names example revenue codes to start with
The table looks right but no files are written	`AUTOTASKS` / `MWL_SPOOL_PATH` is blank or names a directory that does not exist. The `EmitWorklistFiles` log line says which
Files are written but a scanner sees nothing	The two halves are pointed at different directories, or the DICOM server is not running. Run `Test-Mwl.ps1` on the server; the **Mp10 Modality Worklist** guide covers this in full
A study is on the table but has no file	It has no modality — see `no modality (not emitted)` in the log. A laboratory or office-visit order belongs in that count

The file check does not complain when a table cannot be created. If the `modalitywl` table is missing after a check that appeared to finish normally, do not assume the setting is wrong — the check can fail silently on a structure problem. Report it rather than re-running it repeatedly. *(IT task.)*

18.10 Appointment reminders are not going out

Symptom	Likely cause
No texts at all	`SMS` / `SEND_APPOINTMENT_REMINDERS` is not `YES`, or `APPOINTMENT_REMINDERS_ALERT_DAYS` is `0`
Queued but never sent	The `TWILIO` account settings — check `ACCOUNT_SID`, `AUTH_TOKEN` and `TEL_FROM`
Texts stop partway through the day	`MAX-DAILY-SMS` reached. The log says so explicitly
A patient with two appointments got one text	Working as designed — one reminder per patient per day
An appointment just entered was skipped	It is inside the `COOLING_PERIOD` — it is texted on a later cycle
Nothing for a specific patient	No mobile number on the appointment, a number under 10 digits, no time on the appointment, or it is already confirmed

18.11 A scheduled report did not arrive

Symptom	Likely cause
It never ran at all	Check **Suspend** is not ticked; check the frequency matches today
Set up today, did not run today	The scheduled time had already passed when you saved it — it starts tomorrow
Ran, but nobody received it	The `eMail` settings, most often an expired or non-app-specific password
One person did not get it	Their address on their user record, not on the report

Symptom	Likely cause
Arrived empty	The report's own parameters — run it manually with **Run** and see

18.12 Getting help

When contacting support, having these ready will save a round trip:

- the relevant section of `AutoTasks.out`, covering the time in question
- `python_debug.log`, for remittance PDF problems
- `backup_zip.log` and `backup_zip.stdout.log`, for backup archive problems
- the exact Section and Entry values you have set
- for a remittance problem, the PDF or the 835 file itself

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